Samuel Alexander

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Education

- **Graphic Era Hill University** 2022 - present
  B.tech in Computer Science and Engineering *CGPA/Percentage: 7.58*
- **Carman School** 2022
  12th, Indian Certificate of Secondary Education *CGPA/Percentage: 82%*
- **Carman School** 2020
  10th, Indian Certificate of Secondary Education *CGPA/Percentage: 84%*

Experience

- **VideoDB** 23 April 2026 – Present
  Engineering Intern
  - **Backend APIs and Databases**: Developed a CRM backend using **NestJS (Node.js), TypeScript, and PostgreSQL**; implemented REST APIs for lead ingestion, account management, and campaigns, with input validation, API-key authentication, and transactional writes.
  - **Background Jobs and Reliability**: Implemented idempotent provider execution, transactional outbox recovery, and timezone-aware campaign scheduling; handled duplicate webhooks and protected stored provider credentials with per-user encryption.
  - **Python Services and Automation**: Built a **Python/FastAPI** backend for workflow recording and replay, exposing APIs for skills, compilations, and runs; added durable background processing, queued execution, checkpoints, and resume/cancel support.
  - **Search Pipelines and Performance Evaluation**: Built resumable retrieval benchmarks across six datasets and a **100-hour corpus with 11,369 queries**, tracking latency and failures. BGE Gemma reranking improved VideoDB Robo Hit@1 from **9.74% to 14.63%**.
  - **Data Processing and Validation**: Built export adapters for **six robotics dataset formats**; added shared batch writers, native-reader validation, bounded download retries, and rejection of incomplete media, with regression tests and conversion-resource benchmarks.
  - **Integrations and Deployment**: Integrated Smartlead, HeyReach, and Call.md through APIs and webhooks; deployed the CRM backend on **Azure VMs** and frontend on Vercel, and maintained deployment runbooks, pagination, and operational recovery workflows.

Technical Skills

- **Languages**: Python, TypeScript, JavaScript, SQL, C++
- **Backend**: Node.js, NestJS, FastAPI, REST APIs, Webhooks, Background Jobs
- **Data**: PostgreSQL, Redis, Drizzle ORM, FAISS
- **Engineering Tools**: Git, Docker, Azure VMs, Vercel, Automated Testing